Math 242, S16
Quiz 8

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find the solution of the following initial value problem:

$$y'' + 8y' + 25y = 0, \quad y(0) = 1, \quad y'(0) = -1.$$

Characteristic equation: $r^2+8r+25=0$

$$r = \frac{-8\pm\sqrt{8^2-4\times1\times25}}{2} = \frac{-8\pm\sqrt{-36}}{2} = -4\pm3i$$

$$\Rightarrow y(x) = e^{-4x}(C_1\cos3x + C_2\sin3x)$$

$$y'(x) = -4e^{-4x}(C_1\cos3x + C_2\sin3x) + e^{-4x}(-3C_1\sin3x + 3C_2\cos3x)$$

$$\left.\begin{aligned} y(0)=1 &\Rightarrow C_1 = 1 \\ y'(0)=-1 &\Rightarrow -4C_1+3C_2=-1 \end{aligned}\right\} \Rightarrow \begin{aligned} C_1 &= 1 \\ C_2 &= 1 \end{aligned}$$

Thus $y(x) = e^{-4x}(\cos3x + \sin3x)$

Problem 2. (5 points) Find the general solution of the homogeneous differential equations

$$9y^{(3)} + 12y'' + 4y' = 0.$$

Characteristic equation

$$9r^3+12r^2+4r=0$$

$$r(9r^2+12r+4)=0$$

$$r(3r+2)^2=0$$

$$\Rightarrow r_1=0, \quad r_2=r_3=-\frac{2}{3}$$

$$\Rightarrow y(x) = C_1 + e^{-\frac{2}{3}x}(C_2x+C_3)$$